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RESEARCH PAPER

A Transformer-based approach for anomaly detection in intelligent well completions

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Abstract: This study introduces a novel methodology and makes case studies for anomaly detection in multivariate oil production time-series data, utilizing a supervised Transformer algorithm to identify spurious events related to interval control valves (ICVs) in intelligent well completions (IWC). Transformer algorithms present significant advantages in time-series anomaly detection, primarily due to their ability to handle data drift and capture complex patterns effectively. Their self-attention mechanism allows these models to adapt to shifts in data distribution over time, ensuring resilience against changes that can occur in time-series data. Additionally, Transformers excel at identifying intricate temporal dependencies and long-range interactions, which are often challenging for traditional models. Field tests conducted in the ultradeep water subsea wells of the Santos Basin further validate the model's capability for early anomaly identification of ICVs, minimizing non-productive time and safeguarding well integrity. The model achieved an accuracy of 0.954 4, a balanced accuracy of 0.969 4 and an F1-Score of 0.957 4, representing significant improvements over previous literature models.

Key words: anomaly detection; intelligent well completion; interval control valve; well integrity; oil well monitoring; deep learning; Transformer algorithm

Introduction

In the initial development phases of the Pre-Salt Cluster in the Santos Basin, intelligent well completions (IWC) emerged as a crucial design for Production & Development Projects. This technology enhances reservoir management and addresses uncertainties, particularly vital in carbonate reservoirs. The complexity associated with IWC poses a challenge for well engineering teams, especially in ultra-deepwater, where minimizing construction cost is paramount. Successful and efficient IWC deployment becomes pivotal for the budgetary success of projects, with over 200 IWCs deployed in the last 15 years [1–2].

The typical IWC design for pre-salt reservoirs involves a 273.1 mm×250.8 mm (10 ¾" × 9 ⅞") production casing combination. Perforation operations are carried out before running the completion string in-hole (RIH) using techniques such as tubing-conveyed perforation (TCP) or wireline-conveyed perforation [1–2]. In more recent wells, open-hole completions were employed alongside multi-position, hydraulically controlled interval control valves (ICVs). These ICVs enable remote control of production or injection across 2–3 zones. The valves are operated via 3 or 4 dedicated control lines—one of which is common to all valves, known as the common close line, while the remaining lines are assigned to open individual valves. Given the shared control line, a specific operational logic must be adhered to prevent undesired valve movements. Each interval and the completion string are equipped with dedicated pressure and temperature (P&T) sensors, along with downhole chemical injection systems. A multiplex-controlled wet Christmas Tree (WCT) interfaces with the downhole control lines, providing power and communication to the P&T sensors, thus ensuring efficient monitoring and control of well operations.

During the production phase, some anomalies and failures have been detected in IWC systems. Common issues include undesired movements of ICVs, valve cycling difficulties, hydraulic module malfunctions, scale occurrence risks, stuck or leaking ICVs, and gas migration through control lines causing blockages due to hydrate formation. All these issues with ICVs during production result in a loss of oil production and an increase in in-

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tervention costs with drilling rigs for workover operations [2]. While digital twins effectively monitor real-time well integrity by capturing thermo-mechanical behaviors and loads on well components, they fall short in addressing all potential subsurface or control system-related anomalies such as equipment malfunctions or spurious valve closures [3–4]. For offshore wells, early detection of anomalies donwhole is particularly crucial due to the high costs associated with rig interventions. Thus, it is necessary to introduce a machine learning technique for proactively predicting these anomalies, thereby minimizing production loss and ensuring well integrity.

Recent advancements focus on reconstruction-based time-series anomaly detection methods, which transform time-series data into a latent vector and reconstruct it to detect anomalies through high reconstruction errors. Malhotra et al. [5] used autoencoders for learning the distribution of normal data, while Geiger et al. advanced this approach with TadGAN [6], leveraging generative adversarial network (GANs) for time-series anomaly detection. Other scholars have deployed K-Nearest Neighbors (K-NN) and floating window algorithms to detect issues such as downhole safety valve (DHSV) closure, production loss, line obstructions and control valve closure, and demonstrated the feasibility of these methods in practical applications, which, however, are found with the complexities of ICV failures [7]. Other scholars performed condition-based monitoring (CBM) using Random Forest and Bayesian models, and classified eight operational problems with over 94% accuracy. However, these models exhibit minimal detection delays that may not suffice in scenarios requiring immediate response to ICV malfunctions, emphasizing a gap in timely mitigation for complex failures [8]. Some studies highlighted decision trees as effective in classifying failures, yet they fall short in accurately detecting spurious DHSV closures. An anomaly detection model utilizing a Long Short-Term Memory (LSTM) autoencoder and Support Vector Machines (SVM) classifier showed improved performance for faults with slow dynamics, but these approaches may not adequately address the multifaceted nature of ICV failures within IWCs [9–10]. Recent evaluations of techniques such as Isolation Forest, One-class Support Vector Machine (OCSVM), Local Outlier Factor (LOF), Elliptical Envelope and Autoencoder (both feedforward and LSTM architectures) showcased the Local Outlier Factor as the top performer in various scenarios [11]. While these methods demonstrated competitive accuracy for general anomaly detection, they did not specifically target ICV problems in IWCs [12]. Additionally, the unsupervised machine learning algorithms, particularly the LOF, have shown competitive accuracy in identifying spurious DHSV closures, but they have not successfully identified ICV issues within IWCs [13]. A recent study employed a combination of deep learning autoencoder and a rule-based approach for anomaly detection, which demonstrated potential in enhancing operational safety and reducing costs associated with non-productive time (NPT) and repairs. However, it failed to capture the effects of malfunctioning ICVs in the context of reservoir zones in intelligent well completions [14].

In contrast, the Transformer method, originally developed for natural language processing, has proven exceptionally effective in sequential data analysis across various domains [15–16]. Its superior performance in long sequence time-series forecasting positions it as a promising tool for addressing the complexities inherent in anomaly detection [17–19]. Transformer algorithms based on self-attention mechanism allow these models to adapt to shifts in data distribution over time, ensuring resilience against changes that may occur in time-series data. Additionally, Transformers excel at identifying intricate temporal dependencies and long-range interactions, which are often challenging for traditional models. Unlike recurrent neural networks (RNNs), Transformers can process entire sequences in parallel, leading to faster training and inference times, making them suitable for large datasets. Their flexibility in input representation accommodates various feature types, enhancing the modeling of complex time-series data. Furthermore, the rich contextual information that Transformer gather through their attention mechanism aids in accurately identifying anomalies that may otherwise go unnoticed, making them a powerful tool for detecting unusual patterns and behaviors in time-series data.

This research presents a Transformer-based approach for time series anomaly detection, targeting malfunction events in ICVs within IWCs, and verifies it by a case study on pre-salt reservoirs in the Santos Basin.

1. Methodology

1.1. Transformer model for time series

We propose a time-series anomaly detection method based on a Transformer algorithm and construct a corresponding time series prediction model, as shown in Fig. 1. The model comprises an encoder that learns a hierarchical representation from multiple Transformer encoder layers and a decoder that integrates this representation. The goal is to minimize the error between predicted and actual sequences, thereby learning the distribution of normal data prevalent in the training set.

In time-series data, each timestamp's input (e.g., temperature, pressure, valves sensor readings) has dependencies on past and future timestamps. The self-attention mechanism allows the model to compute these dependencies flexibly. The self-attention mechanism is a key

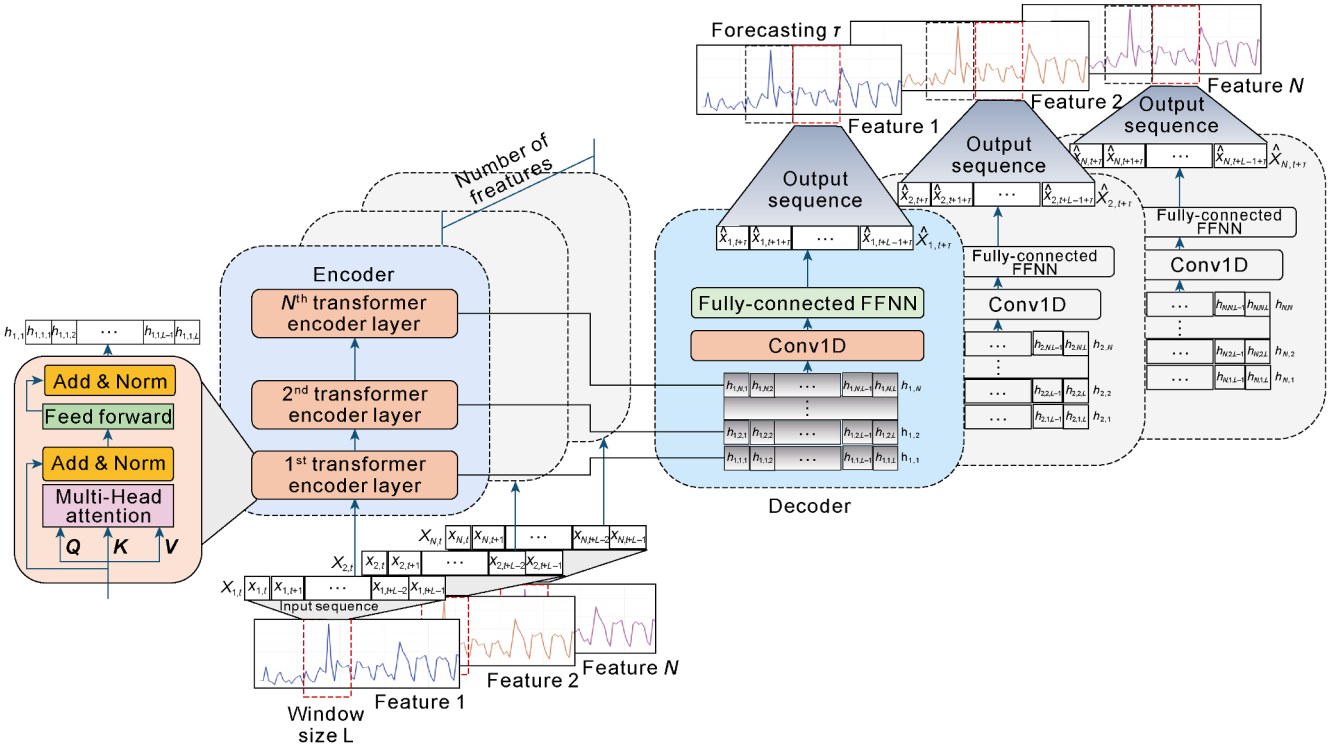


Fig. 1. Architecture of the proposed Transformer model.

feature of the Transformer. Multiple vectors, each representing a token, exist in the input sequence. For each input vector, three different linear projections are created, i.e. Query (***Q***), Key (***K***) and Value (***V***). These projections are learned during training. For each query, its compatibility with all keys is computed as a dot product, so that a matrix of Attention Score is obtained. The scores are normalized using a Softmax function to obtain attention weights. These weights indicate how much attention each token should pay to others. Using the attention weights, a weighted sum of the value vectors is computed for each query. This results in a new representation of the sequence.

Multi-head attention extends the self-attention mechanism by allowing the model to focus on different aspects of the time series data simultaneously. Each "head" in the multi-head attention mechanism learns to capture different types of temporal relationships or patterns within the data. For example, while some heads may attend to long-term trends, others may focus on short-term fluctuations or seasonal patterns. Each head operates independently with different learned projection matrices for queries, keys and values. The outputs from these heads are then concatenated and linearly transformed, resulting in a richer representation of the time series.

To prepare the time-series data for continuous prediction, the input is segmented into windows of length L. These windows are created using a sliding mechanism with a step size of 1. For Feature 1, as an example, the window $X_{1,t}=\{x_{1,t}, x_{1,t+1}, \ldots, x_{1,t+L-1}\}$is obtained, starting at time t.

Each window undergoes processing through N Transformer encoder layers. For example, the window $X_{1,t}$ for Feature 1 is transformed to $h_{1,1}=\{h_{1,1,1}, h_{1,1,2}, \ldots, h_{1,1,L}\}$ after processing at the 1st Transformer encoder layer, and then to $h_{1,2}=\{h_{1,2,1}, h_{1,2,2}, \ldots, h_{1,2,L}\}$ after processing at the 2nd Transformer encoder layer; and the like, finally to $h_{1,N}=\{h_{1,N,1}, h_{1,N,2}, \ldots, h_{1,N,L}\}$ after processing at the Nth Transformer encoder layer.

Each layer employs masked multi-head self-attention to capture dependencies within the sequence, ensuring no future information leakage by masking the upper triangular elements of the attention matrix. To effectively capture features relevant to tasks such as identifying ICV failure time-series characteristics, the features learned across the Transformer layers are aggregated. For this purpose, the representations (e.g. $h_{1,1}$, $h_{1,2}$, …, $h_{1,N}$ for Feature 1) are stacked to form the final encoder output. This approach integrates both local details from lower layers and global dependencies from higher layers within the sequence.

The decoder receives the final encoder output and utilizes a 1-dimensional convolutional neural network (1D CNN) to combine hierarchical information. The 1D CNN efficiently captures time-invariant characteristics with fewer parameters compared to traditional RNNs, enhancing its suitability for time-series modeling. From the 1D CNN output, a fully-connected layer generates predictions, such as $\hat{X}_{1,t+\tau}$ for Feature 1.

This comprehensive connection between the encoding and decoding processes, alongside the attention mechanisms, enhances the model's ability to capture and pre-

dict ICV failure time-series features, ultimately improving its effectiveness in anomaly detection tasks.

1.2. Training process and anomaly detection

In this study, the prediction model is trained using the mean squared error (MSE) as the loss function, which measures the square errors between the actual values (windows) and the predicted values (outputs) and normalize them by the sequence length L and the dimensionality d. After training, the prediction model learns to approximate the distribution of normal data from the training set. The loss functions of Features 1–N are summed and then divided by N, to get the total loss:

To detect anomalies in a real time manner, the prediction error for each timestamp is quantified as the anomaly score. This score is defined by the Euclidean distance between the actual values and the predicted values across all dimensions.

During the inference phase, the model processes the most recent L timestamps $X_{t-L+1}=\{x_{t-L+1}, x_{t-L+2}, \ldots, x_t\}$ as input and predicts the sequence τ time steps into the future $X_{t-L+1+\tau}=\{x_{t-L+1+\tau}, x_{t-L+2+\tau}, \ldots, x_{t+\tau}\}$. The anomaly score is calculated in real-time based on the last τ predicted values. To determine if a data point is anomalous, a threshold T is set based on the mean μ and standard deviation σ of the reconstruction errors, as expressed by:

$$T = \mu + k\sigma \tag{1}$$

k is a parameter that determines the tolerance interval around the mean. The acceptance criteria, which are based on tolerance intervals around the mean, can be applied to both sequential and non-sequential datasets, as this method aims to encompass a specified proportion of the population with a defined level of confidence. Employing the properties of the normal distribution in this manner is a well-established and widely recognized approach in the scientific domain of anomaly detection [20–21]. By establishing a threshold using the mean plus three standard deviations (k=3), any data point that lies outside the central 99.7% range can be identified as an anomaly. An instance is flagged as anomalous if its anomaly score exceeds T. This methodology allows for effective real-time anomaly detection by leveraging the trained prediction model to continuously monitor and evaluate incoming time-series data against learned normal patterns. Despite its simplicity, it yielded consistent results and does not impose computational overhead on real-time monitoring.

One characteristic that Transformer models enable us easy to evaluate, beyond available attributes, is the use of calculated features, such as the pressure differential between the pressure downhole gauges (PDGs) in the annulus, as well as the PDGs within the column. These variations can be correlated with changes in flow, indicating a shift in the position or closure of one or more ICVs. In contrast, analyzing sensor data as a separate feature could not capture this physical phenomenon effectively.

1.3. Metrics for performance evaluation

In evaluating the performance of classification algorithms, several key metrics play pivotal roles. These metrics include Accuracy (ACC), Precision (PR), Recall (REC), Specificity (SP), F1 Score, balanced accuracy (ACCb) and Instances Identification Percentage (IIP).

Accuracy (ACC) is fundamental, representing the proportion of correctly classified samples—both normal and fault—out of the total dataset. It is calculated using the sum of true positives (TP) and true negatives (TN) divided by the total number of samples. TP and TN are the correctly identified instances of normal and fault conditions, respectively. Precision (PR) assesses the proportion of true positives among all predicted positives, helping to gauge the exactness of the algorithm in identifying faults. It is determined by the TP divided by the sum of TP and false positives (FP). FP denotes instances incorrectly flagged as faults. Recall (REC) measures the ratio of true positives to all actual positives, and it is crucial in industrial applications. It is calculated using the TP divided by the sum of TP and false negatives (FN). FN represents fault instances incorrectly classified as normal. Specificity (SP) evaluates the algorithm's ability to correctly identify true negatives among all actual negatives, complementing the evaluation of false positives. It is determined using the TN divided by the sum of TN and FP. The F1 Score, a harmonic mean of Precision and Recall, offers a balanced measure especially useful in scenarios with class imbalance. For cases of unbalanced data, where accuracy might not reflect true performance, balanced accuracy (ACCb) provides a more suitable alternative. It is the arithmetic mean of Recall and Specificity. Additionally, the Instances Identification Percentage (IIP) measures the classifier's ability to correctly identify instances of a specific class in the test set. A value of 1.0 indicates perfect identification across all instances tested.

1.4. Model parameterization

In this study, the performance of the predictive model was optimized using public datasets and real-world data from monitoring system. The focus was on exploring various hyperparameter combinations without exhaustively testing every possibility due to computational constraints. The study leveraged Python 3.9 for implementation and employed a MinMax scaler to normalize the data.

For the public datasets, a training window and a forecasting window were defined firstly. The training window consisted of 60 samples for each attribute, with a time interval of 60 s between samples. Subsequently, the forecasting window spanned the next 60 s. This setup

Table 1. Hyperparameters used for model training and tests for different databases

Type of dataset	Number of layers	Number of input features	Number of heads	L	τ	Learning rate	Epochs
Public dataset	16	3	3	60	60	0.001	20
Real-world data	16	4–12	4–12	60	60	0.001	10

allowed us to evaluate model performance under controlled conditions with well-defined intervals.

In contrast, the real-world data from the monitoring system arrived at a lower frequency, with data points collected every 30 s. Consequently, the training period comprised 60 samples per attribute, corresponding to a 30-min window. The subsequent test period utilized 60 samples to evaluate performance over the subsequent 30 min. This approach accommodated the inherent differences in data frequency and temporal dynamics compared to the public datasets.

For determining the Number of Layers and Number of Heads, a grid search method was employed. In contrast, the selection of other hyperparameters, such as learning rates and batch sizes, was guided by considerations of computational feasibility and the loss criteria for the validation dataset. Each parameter combination was evaluated based on performance metrics, including accuracy and F1-Score. Table 1 presents the hyperparameters utilized in the public datasets and the specific cases studied.

2. Case studies

Utilizing hyperparameters, training processes, and anomaly detection as presented above, the model performance was initially evaluated based on the public dataset [12] and compared to existing literature models. Subsequently, the model was expanded to wells with IWCs in the Pre-Salt Cluster of the Santos Basin, aiming to demonstrate the effectiveness of the Transformer model in complex scenarios.

2.1. Model performance evaluation based on public dataset

There is a lack of public data available for events related to IWCs and ICVs. To initially train and validate our methodology, we utilized well data from the public database [12] under normal production conditions (class 0 in the public base). Subsequently, this model was compared with other techniques, focusing on spurious DHSV closure events (class 2 in the public dataset) obtained from the public dataset. Automatic detection of these closures can enable timely corrective actions, preventing production losses and additional costs. Key attributes used include those acquired by the Temperature/Pressure Transducer (TPT) of the Wet Christmas Tree and the PDG, as presented in Table 2. The locations of TPT, PDG and DHSV are shown in Fig. 2. Data in the public database were acquired at one-second intervals. The transient and steady states of failure were unified as a

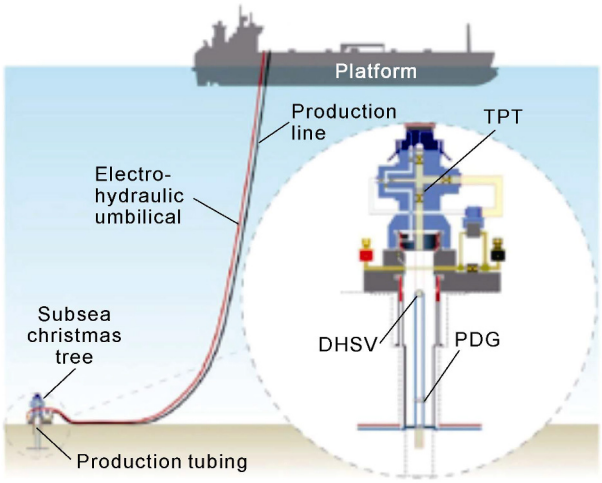


Fig. 2. Schematic view of the offshore well connected to the production platform [12].

Table 2. Available attributes in the public database

Attribute	Description	Unit
P-TPT	Pressure on TPT	Pa
T-TPT	Temperature on TPT	°C
P-PDG	Pressure on PDG	Pa

single anomalous state, that is, a binary classification (normal and failure) was performed. There is no not-a-number (NaN) value in the variable monitoring data. However, there are few data points without labels (NaN values in the class column). For these samples, the data located between the transient and steady-state periods of failure were considered a failure, and the rest as normal.

The model proposed in this study was aligned with previous research on similar anomalies, using methodologies like Random Forest (RF) [8], Decision Tree (DT) [9], LSTM autoencoder [10], Decision Diagram (DD)+LSTM [14], and LOF [13]. Notably, the DD+LSTM [14] and LOF [13] methods showed satisfactory results for spurious DHSV closure events.

Table 3 summarizes the performance evaluation results of the Transformer model proposed herein for spurious DHSV closure events in the public database [12], in terms of ACC, F1-Score and ACCb metrics. Table 4 presents the comparison between five literature classifiers and the Transformer model, in terms of ACC, F1-Score and IIP metrics. Given the same dataset, the Accuracy achieved by the Transformer model is around 0.986, in contrast to 0.999 by LSTM autoencoder and LOF, 0.989 by DD+LSTM, 0.871 by RF and 0.600 by DT. From F1-Score values, LOF and DD+LSTM perform the best, followed by the Transformer model and LSTM model, both of which are significantly superior to DT. The IIP

Table 3. Summary of identification results of the Transformer model for spurious DHSV closure events from public database

Dataset ID	Name of dataset	ACC	F1-Score	ACCb
1	WELL-00002_20131104014101.csv	0.993 2	0.994 2	0.986 1
2	WELL-00003_20141122214325.csv	0.993 6	0.995 7	0.983 2
3	WELL-00003_20170728150240.csv	0.991 2	0.993 4	0.980 6
4	WELL-00003_20180206182917.csv	0.987 9	0.993 2	0.942 0
5	WELL-00009_20170313160804.csv	0.991 3	0.990 9	0.982 7
7	WELL-00011_20140515110134.csv	0.971 6	0.985 5	0.950 0
8	WELL-00011_20140530100015.csv	0.969 5	0.980 2	0.937 7
10	WELL-00011_20140720120102.csv	0.994 9	0.993 9	0.995 7
11	WELL-00011_20140726180015.csv	0.982 5	0.988 3	0.966 6
13	WELL-00011_20140916060300.csv	0.997 1	0.995 8	0.997 8
14	WELL-00011_20140921200031.csv	0.981 6	0.987 7	0.964 0
15	WELL-00011_20140928100056.csv	0.989 2	0.985 2	0.991 5
16	WELL-00011_20140929170028.csv	0.977 7	0.982 8	0.969 2
17	WELL-00011_20140929220121.csv	0.991 7	0.984 4	0.994 4
18	WELL-00011_20141005170056.csv	0.993 9	0.994 7	0.992 9
19	WELL-00011_20141006160121.csv	0.989 3	0.986 5	0.991 2
20	WELL-00012_20170320033022.csv	0.964 7	0.968 6	0.961 1
21	WELL-00012_20170320143144.csv	0.982 0	0.980 5	0.983 5
22	WELL-00013_20170329020229.csv	0.994 2	0.995 4	0.992 1
mean		0.986 1	0.988 2	0.976 9

Table 4. Comparison of identification results between literature classifiers and the Transformer model for spurious DHSV closure events from public database

Classifier	ACC	F1-Score	IIP
RF	0.870 8		0.58
DT	0.600 0	0.490 0	
LSTM	0.999 2	0.936 0	1.0
DD+LSTM	0.989 4	0.991 7	1.0
LOF	0.999 1	0.996 9	1.0
Transformer	0.986 1	0.988 2	1.0

values indicate that the Transformer, LOF, DD+LSTM and LSTM models successfully identified both normal and failure instances, in contrast to the RF results for this metric.

2.2. Application in ultradeep wells in the offshore pre-salt reservoirs of Brazil

Following the Transformer model's evaluation using a public database, it was deployed in subsea production and injection wells in offshore pre-salt reservoirs in Santos Basin, Brazil, which were equipped with multiplexed WCTs and IWCs [22]. For training, the 3-month production data from each well was used, focusing on periods 2 h before and after well openings, closings, or changes in production/injection flow rates under normal conditions. Extended steady-state production periods were excluded to prioritize capturing anomalies during valve operations in ICVs. Separate training sets were prepared for production and injection wells due to their distinct flow regimes and feature behaviors. The training utilized 80% of the data for minimizing loss, with 20% reserved for validation. Table 5 presents key attributes available in IWC wells, including readings of WCT's TPT, readings of PDGs, and surface pressure and choke position in the production/injection line, the same as those used in previous model studies and real-time monitoring practices [23–24]. The data were recorded at 30-s intervals.

Pressure differential values were calculated between the following sensors shown in Table 5: P-TPT and P-PDG, P-PDG and P-L-PDG, P-PDG and P-U-PDG, P-L-PDG and P-U-PDG. In cases where the well has three producing/injection zones, pressure differentials were also calculated for P-PDG and P-I-PDG, P-L-PDG and P-I-PDG, and P-I-PDG and P-U-PDG. Table 6 presents the calculated attributes available.

The pressure differentials calculated between PDGs in the annulus and between PDGs in the string can provide valuable insights by demonstrating how differential pressures fluctuate, or remain stable, across various reservoir zones. These fluctuations can be linked to changes in flow, suggesting possible movements or closures of one or more ICVs. Additionally, this approach may be more effective for use in machine learning models, as it could be less sensitive to noise originating from the sensors.

Given that the Transformer model allows for the use of multiple features, several model configurations were evaluated: (1) TRANS-A, which considered all sensors presented in Table 5 (total of 12 features); (2) TRANS-B, which included only the data from the TPT of WCT and the bottom hole PDGs (total of 4 or 5 features, depending

Table 5. Available attributes in IWC wells

Attribute	Description	Unit
P-TPT	Pressure on TPT	Pa
T-TPT	Temperature on TPT	ºC
P-PDG	Pressure on PDG	Pa
P-U-PDG	Pressure in front Upper Zone - PDG	Pa
P-I-PDG	Pressure in front Intermediate Zone - PDG	Pa
P-L-PDG	Pressure in front Lower Zone - PDG	Pa
T-PDG	Temperature on PDG	°C
T-U-PDG	Temperature in front Upper Zone - PDG	°C
T-I-PDG	Temperature in front Intermediate Zone - PDG	°C
T-L-PDG	Temperature in front Lower Zone - PDG	°C
P-SURFACE	Surface pressure in the well production/injection line	Pa
CHOKE	Choke position in the well production/injection line	%

Table 6. Calculated attributes in IWC wells

Attribute	Description	Unit
DP1	Pressure differential between TPT and PDG	Pa
DP2	Pressure differential between PDG and Upper Zone - PDG	Pa
DP3	Pressure differential between PDG and Lower Zone - PDG	Pa
DP4	Pressure differential between Upper Zone – PDG and Lower Zone - PDG	Pa
DP5	Pressure differential between PDG and Intermediate Zone - PDG	Pa
DP6	Pressure differential between Lower Zone – PDG and Intermediate Zone - PDG	Pa
DP7	Pressure differential between Intermediate Zone - PDG and Upper Zone - PDG	Pa

on whether the well has two or three zones); (3) TRANS-C, which included the pressure sensor data and the calculated pressure differential data in Table 6 (8 or 12 features, depending on whether the well has two or three zones); and (4) TRANS-D, which considered only the calculated pressure differential presented in Table 6 (4 or 7 features, depending on whether the well has two or three zones).

An analysis was conducted on the monitoring data during production in pre-salt reservoirs in the Santos Basin from late 2022 to 2024, obtaining eight case studies from six wells in the basin to validate the effectiveness and adaptability of the Transformer model in ICV fault detection. These case studies include: (1) Well #1: An oil production well in ultradeep water (2 119 m), drilled vertically to a depth of 5 243 m, equipped with ICVs, and produces from 4 898 m to 5 129 m; (2) Well #2: A water injection well in ultradeep water (2 128 m), drilled vertically to 5 275 m, also equipped with ICVs, and injects water into zones from 4 917 m to 5 180 m; (3) Well #3: A water alternating gas (WAG) injection well in ultradeep water (1 900 m), drilled directionally to 6 111 m, using ICVs for injection across zones from 5 585 m to 5 895 m; (4) Well #4: A WAG injection well in ultradeep water (2 027 m), drilled vertically to 5 933 m, equipped with ICVs, and injects water into target zones from 5 591 m to 5 871 m; (5) Well #5: An oil production well in ultradeep water (2 125 m), open hole completion, drilled directionally to 5 343 m, equipped with ICVs, and produces from 5 152 m to 5 255 m; (6) Well #6: A WAG injection well in ultradeep water (2 183 m), open hole completion, drilled directionally to 5 538 m, using ICVs for injection into zones from 5 362 m to 5 530 m. Wells #1, 2#, 5# and 6# have two injection/production zones, while wells #3 and 4# have three injection/production zones. The diverse conditions presented by these wells highlight the Transformer model's robustness in varying depths, trajectories and operational configurations. The case studies reveal that four anomalies were detected in production well scenarios (wells #1 and #5), while the remaining four spurious anomalies were identified in injection well scenarios (wells #2, #3, #4, and #6).

The models with four configurations were applied to eight case studies. As illustrated in Table 7, the metrics of TRANS-D are superior to the other three models. This may be attributed to the fact that TRANS-D uses the calculated pressure differential between PDGs as the input feature, which can yield more significant insights on a shift in the position or closure of ICVs as compared to the case directly using the PDG reading as the input. The results obtained by TRANS-D are presented in Figs. 3–8, and compared with the human post-classification efforts.

Fig. 3 compares the model detection and human post-classification for events in Well #1. As illustrated,

Table 7. Results of the Transformer models for case studies

Case	Model	ACC	ACCb	F1-Score
Well #1, 1st event	TRANS-A	0.951 8	0.970 9	0.927 4
	TRANS-B	0.951 9	0.970 9	0.927 6
	TRANS-C	0.950 8	0.970 2	0.925 8
	TRANS-D	0.964 2	0.979 4	0.947 5
Well #1, 2nd event	TRANS-A	0.914 5	0.923 5	0.924 5
	TRANS-B	0.821 8	0.833 1	0.848 8
	TRANS-C	0.912 2	0.921 3	0.922 5
	TRANS-D	0.962 5	0.970 3	0.969 2
Well #1, 3rd event	TRANS-A	0.937 7	0.959 5	0.940 7
	TRANS-B	0.910 8	0.937 0	0.910 8
	TRANS-C	0.933 3	0.955 8	0.935 7
	TRANS-D	0.970 2	0.979 1	0.978 7
Well #2 event	TRANS-A	0.969 6	0.988 4	0.931 8
	TRANS-B	0.826 5	0.904 6	0.624 4
	TRANS-C	0.826 5	0.904 6	0.624 4
	TRANS-D	0.970 4	0.988 9	0.934 6
Well #3 event	TRANS-A	0.799 7	0.812 5	0.831 8
	TRANS-B	0.800 5	0.813 3	0.832 4
	TRANS-C	0.745 9	0.760 2	0.794 6
	TRANS-D	0.974 0	0.981 6	0.980 6
Well #4 event	TRANS-A	*	*	*
	TRANS-B	0.661 5	0.645 5	0.450 8
	TRANS-C	0.903 9	0.931 6	0.924 3
	TRANS-D	0.902 6	0.930 3	0.923 1
Well #5 event	TRANS-A	0.465 6	0.507 0	0.636 2
	TRANS-B	0.874 4	0.891 8	0.888 5
	TRANS-C	0.891 6	0.908 0	0.903 6
	TRANS-D	0.978 8	0.990 0	0.988 5
Well #6 event	TRANS-A	0.908 3	0.931 4	0.929 5
	TRANS-B	0.900 0	0.923 3	0.921 8
	TRANS-C	0.900 0	0.923 3	0.921 8
	TRANS-D	0.912 8	0.935 8	0.937 5

Note: * means that the model did not identify the anomaly.

TRANS-D can effectively detect anomalies related to the lower ICVs. In the 1st event (Fig. 3a), the model accurately identified an anomaly at 10:29 PM on 09/10/22 based on pressure differentials calculated and predicted. Subsequently, the operator performed ICV cycling to address the issue, with human classification confirming the event at 01:04 AM on 09/11/22. This event demonstrates the model's prompt detection capability, facilitating timely operator intervention to mitigate potential production disruptions. In the 2nd event (Fig. 3b), the model detected an anomaly at 02:17 PM on 02/25/23. The operator initiated ICV cycling to rectify the issue, and human post-classification identified the anomaly at 4:24 PM on the same day. This scenario underscores the model's ability to preemptively signal anomalies, enabling proactive maintenance actions to uphold operational continuity. In the 3rd event (Fig. 3c), the model detected an anomaly at 11:28 PM on 09/29/23, aligning closely with human classification at 10:20 PM. Both identifications confirmed a spurious closure event of the lower ICV. This synchronization highlights the model's consistency in identifying critical anomalies, reinforcing its reliability in real-time monitoring applications. In this specific case, the detection of the anomaly resulted in savings of approximately 400 thousand dollars by preventing a 4 h interruption in

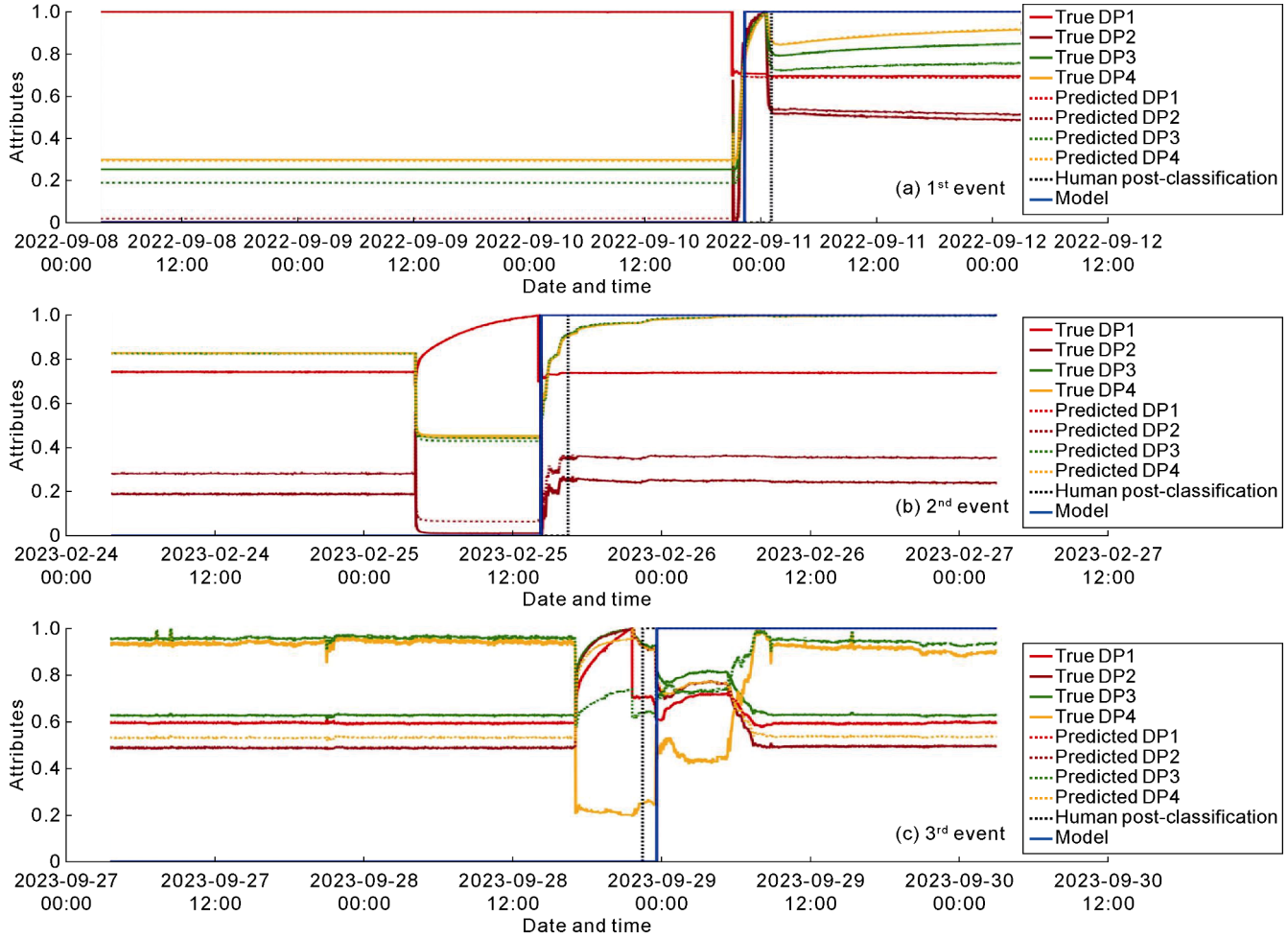


Fig. 3. Anomalous events of TRANS-D detection and human post-classification in Well #1.

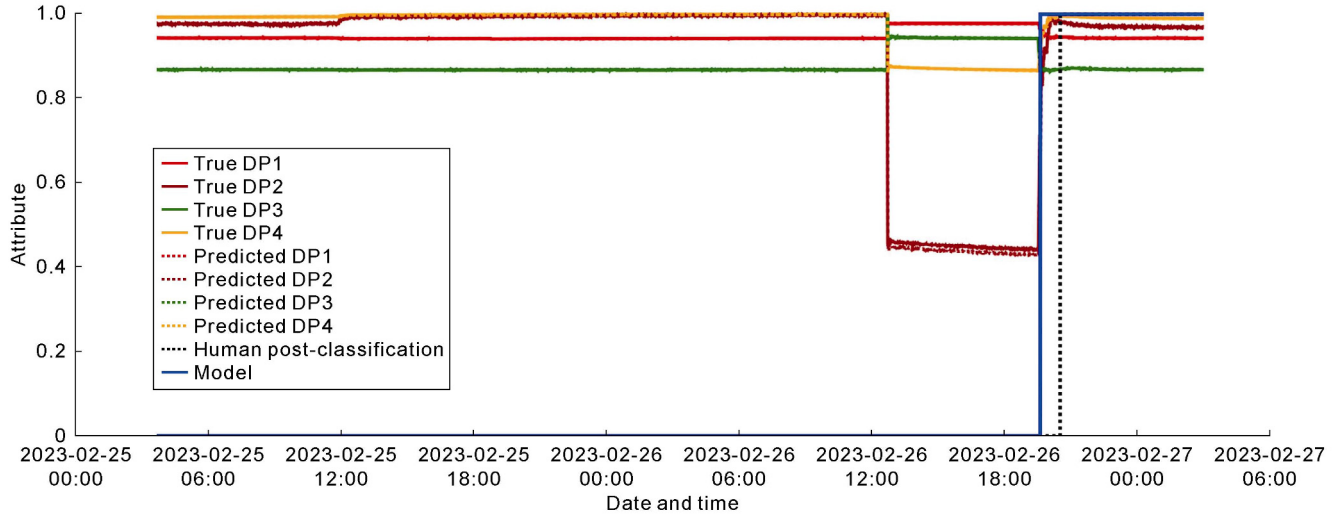


Fig. 4. Anomalous events of TRANS-D detection and human post-classification in Well #2.

production loss at a well with a flow rate of about 3 500 m^3/d.

Fig. 4 depicts anomalous events from 02/25/23 to 02/27/23 in Well #2. The model identified an anomaly at 07:30 PM on 02/26/23, and correctly classified it as a spurious closure event. The human post-classification confirmed the event at 8:28 PM on 02/26/23, prompting the operator to initiate valve cycling and reopening procedures to restore safe production conditions.

Fig. 5 shows the anomalous events in Well #3 from 07/05/23 to 07/08/23. The model detected an anomaly at 3:13 AM on 07/06/23, and identified it as a spurious closure of the lower ICV. The human post-classification verified the event at 4:47 PM on 07/06/23, underscoring

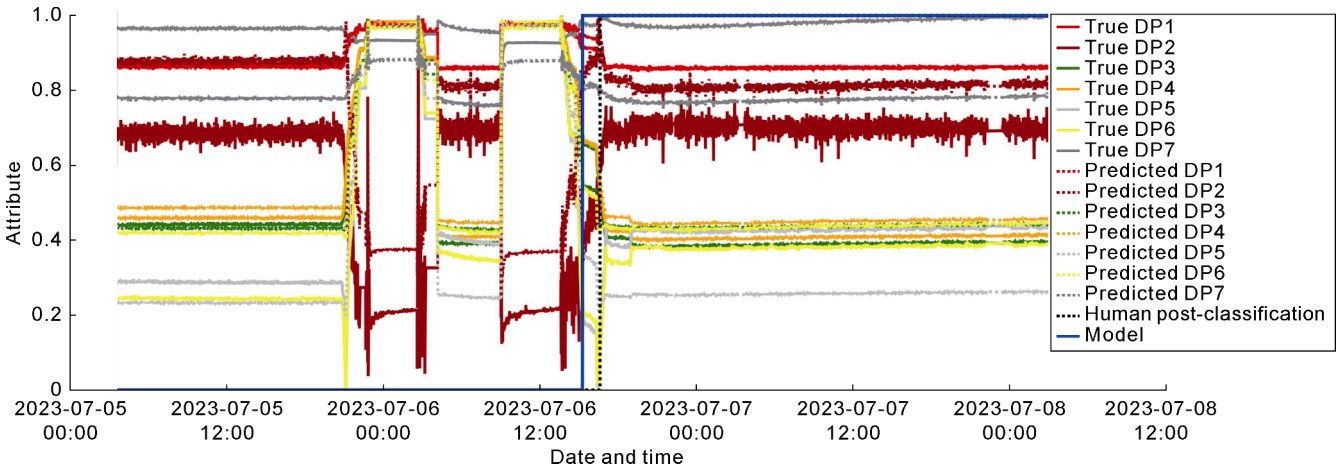


Fig. 5. Anomalous events of TRANS-D detection and human post-classification in Well #3.

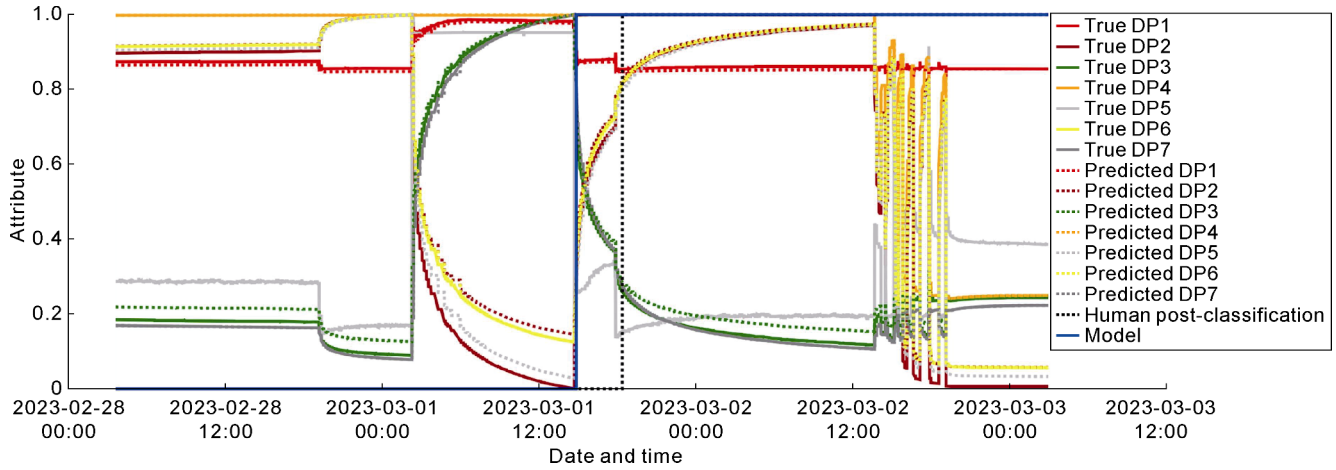


Fig. 6. Anomalous events of TRANS-D detection and human post-classification in Well #4.

the model's early detection capability in operational scenarios.

Fig. 6 presents the anomalous events in Well #4 from 02/28/23 to 03/03/23. The model identified an anomaly at 2:45 PM on 03/01/23, indicating a potential issue with the ICV. The operator subsequently confirmed and addressed the anomaly with valve cycling, and human post-classification marked the event's identification at 6:25 PM on 03/01/23. This case illustrates the effective collaboration between model's automated detection and manual intervention to maintain operational integrity.

Fig. 7 illustrates the anomalous events in Well #5 from 04/17/23 to 04/19/23. The model accurately identified an anomaly at 4:36 AM on 04/18/23, signaling a spurious

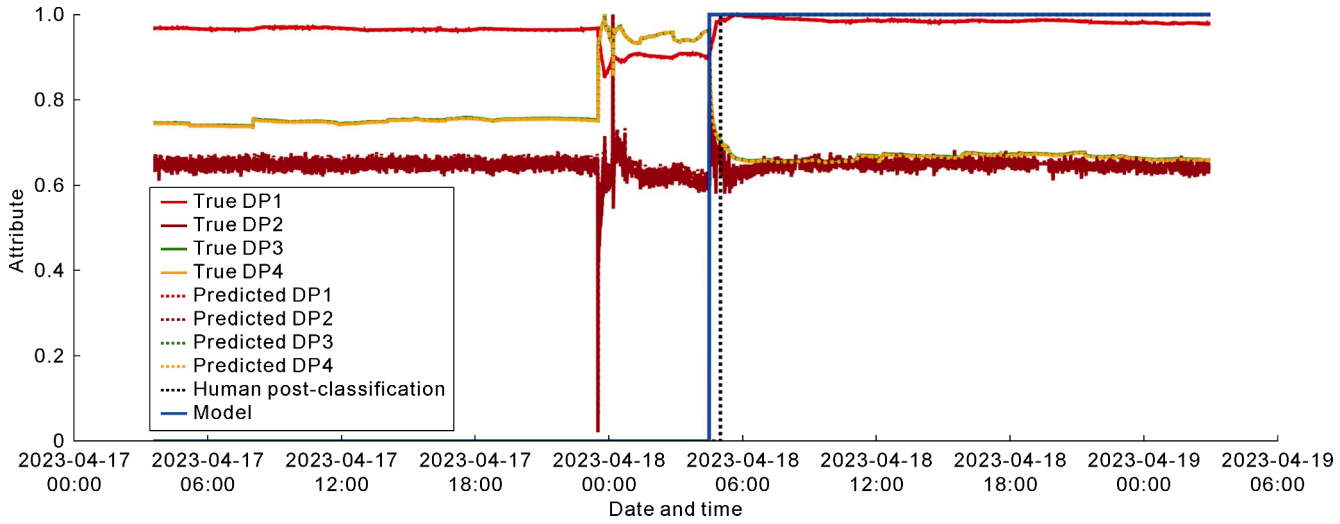


Fig. 7. Anomalous events of TRANS-D detection and human post-classification in Well #5.

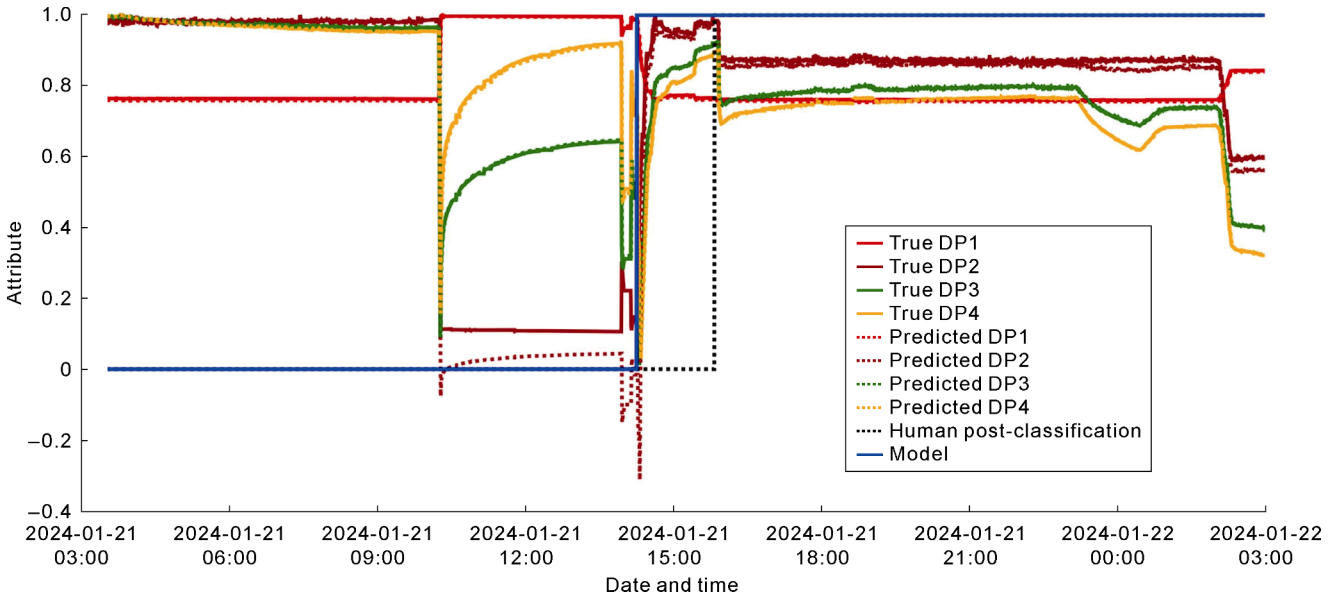


Fig. 8. Anomalous events of TRANS-D detection and human post-classification in Well #6.

closure of ICV. The operator subsequently responded with valve cycling, and human post-classification validated the event at 05:00 AM on 04/18/23. This process ensured swift corrective actions and minimized production disruptions.

Fig. 8 depicts the anomalous events in Well #6. The model identified an event at 2:16 PM on 01/21/24, highlighting a potential ICV anomaly. The operator promptly addressed the issue through valve cycling, and human post-classification confirmed the event at 3:43 PM on the same day. This example underscores the model's reliability in detecting and prompting timely corrective measures to uphold operational efficiency in complex offshore environments.

The analysis of data from various events across wells in the Santos Basin highlights the efficacy of the proposed Transformer model in detecting and classifying anomalous events, particularly spurious closures of ICVs. In Well #1, for instance, three specific events were identified and corrected following the model's detection, with human classification confirming these anomalies as ICV-related spurious closures. Similar successful detections were achieved in Well #2, Well #3, Well #4, Well #5 and Well #6, each illustrating instances where the model correctly identified anomalies ahead of human confirmation and corrective actions by operators. This contributes to well integrity management and helps in better defining operating conditions and extending the well's lifespan [25]. Notably, a comparative study involving DD+LSTM, LOF and TRANS-D models, using pressure differentials as input features, indicated superior performance of the TRANS-D model in detecting ICV spurious closures, achieving detection rates exceeding 0.9 in real-world production scenarios with comprehensive results summarized in Table 8. The TRANS-D performance metrics, such as ACCb of 0.969 4 and an F1-Score of 0.957 4, in predicting ICV closure or partial closure events, indicate the model's efficacy in accurately identifying critical operational anomalies, crucial for maintaining production efficiency and preventing costly downtime where DD+LSTM and LOF models could not yield satisfactory results.

A critical advantage of the fast anomaly detection provided by the Transformer model is the ability for operators to swiftly address issues related to production or injection. The main impacts of ICV closure include reductions in production or injection rates, which, over time, can compromise operational efficiency and well integrity, potentially necessitating costly workover interventions. Without the Transformer model, human detection

Table 8. Comparison of metrics between DD+LSTM, LOF and TRANS-D for IWC wells with ICV spurious movement events

Cases	DD+LSTM			LOF			TRANS-D		
	ACC	ACCb	F1-Score	ACC	ACCb	F1-Score	ACC	ACCb	F1-Score
Well #1, 1st event	0.630 6	0.701 2	0.561 1	0.682 1	0.781 0	0.636 3	0.964 2	0.979 4	0.947 5
Well #1, 2nd event	0.816 7	0.811 9	0.768 6	0.828 6	0.834 1	0.853 9	0.962 5	0.970 3	0.969 2
Well #1, 3rd event	0.762 5	0.732 6	0.769 6	0.706 0	0.753 2	0.733 9	0.970 2	0.979 1	0.978 7
Well #2 event	0.382 9	0.224 2	0	0.308 6	0.596 8	0.295 3	0.970 4	0.988 9	0.934 6
Well #3 event	0.814 3	0.775 3	0.863 8	0.892 7	0.847 2	0.926 7	0.974 0	0.981 6	0.980 6
Well #4 event	0.727 1	0.733 9	0.734 4	0.472 9	0.500 0	0.642 8	0.902 6	0.930 3	0.923 1
Well #5 event	0.669 7	0.689 1	0.739 3	0.484 0	0.510 2	0.649 6	0.978 8	0.990 0	0.988 5
Well #6 event	0.705 0	0.713 0	0.766 7	0.768 1	0.775 7	0.817 9	0.912 8	0.935 8	0.937 5
Mean	0.688 6	0.672 6	0.650 4	0.642 8	0.699 7	0.694 5	0.954 4	0.969 4	0.957 4

of anomalies could take more than a day in some cases, leading to significant economic consequences. The model's performance across various well types and operating conditions in the Santos Basin illustrates its robustness and adaptability, reinforcing its potential as a valuable tool for monitoring and ICV fault detection in complex offshore environments.

Despite demonstrating positive results in the eight anomaly cases studied and being implemented in around 200 wells with IWCs, the model has limitations. It requires a reasonable initial operating period with well opening and closing conditions, as well as a stable operating period, to be used for training. Additionally, the model needs several hours for training with the well's data. The trained model requires specific data from the production or injection well to understand the normal operating conditions. This necessity limits its applicability to other scenarios involving wells with ICVs, given the varying reservoir conditions, fluids and string configurations.

3. Conclusions

This study details the development and implementation of a Transformer model designed to detect anomalies in multivariate time-series data related to oil production. The model targets identifying spurious events involving ICVs in IWC and DHSVs. Testing results indicate that the Transformer model, with pressure differential as the input feature, outperforms previous models (e.g. DD+LSTM and LOF), achieving an accuracy (ACC) of 0.954 4, a balanced accuracy (ACCb) of 0.969 4 and an F1-Score of 0.957 4 for predicting ICV closure or partial closure events. These outcomes highlight the model's precision and reliability. Additionally, the Transformer model exhibits remarkable accuracy in detecting spurious DHSV closure events, reaching an ACC of approximately 0.986 using a publicly available database, comparable to or exceeding other documented models.

For case studies, eight anomalous events were selected from the pre-salt cluster in the Santos Basin. Four anomalies were detected in production wells, and the remaining four in injection wells. This distinction highlights the Transformer model's adaptability to different operational environments, demonstrating its reliability and effectiveness in offshore pre-salt reservoirs. Rapid anomaly detection enables operators to quickly address issues related to production or injection, mitigating the impacts associated with ICV closure. The Transformer model not only enhances operational efficiency but also serves as a critical tool for ensuring the economic viability and safety of offshore oil and gas operations. It can be promoted for large-scale adoption in IWCs and is expected to adapt to other complex well control systems or multivariate fault detection scenarios.

Future research should expand the Transformer-based approach from anomaly detection to anomaly prediction, enhancing its utility in proactive operational management through continuous feedback mechanisms. A key development would be designing a general model that can be trained using data from multiple wells, broadening its applicability. Integrating the Transformer model into digital twin frameworks also represents a promising avenue for enhancing well integrity monitoring. Digital twins provide synthetic sensors for real-time simulation and monitoring, allowing the advanced anomaly detection of the Transformer model to enhance predictive maintenance strategies. This integration would enable continuous adaptation to shifting operational environments, amplifying the model's contribution to sustainable and efficient offshore well operations.

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Nomenclature

$h_{1,1}$, $h_{1,2}$, $h_{1,N}$—new sequences obtained after the window $X_{1,t}$ is processed through the first, second, and N^{th} Transformer encoder layers;

$h_{1,1,1}$, $h_{1,1,2}$, …, $h_{1,1,L}$—data in the sequence $h_{1,1}$;

$h_{1,2,1}$, $h_{1,2,2}$, …, $h_{1,2,L}$—data in the sequence $h_{1,2}$;

$h_{1,N,1}$, $h_{1,N,2}$, …, $h_{1,N,L}$—data in the sequence $h_{1,N}$;

k—a parameter that determines the tolerance interval around the mean;

L—window length;

N—number of features;

$\boldsymbol{Q}$, $\boldsymbol{K}$, $\boldsymbol{V}$—Query, Key, and Value vectors;

t—starting time;

T—threshold;

$x_{1,t}$, $x_{1,t+1}$, …, $x_{1,t+L-1}$—data in the window $X_{1,t}$;

$x_{2,t}$, $x_{2,t+1}$, …, $x_{2,t+L-1}$—data in the window $X_{2,t}$;

$x_{N,t}$, $x_{N,t+1}$, …, $x_{N,t+L-1}$—data in the window $X_{N,t}$;

$\hat{X}_{1,t+\tau}$, $\hat{X}_{1,t+1+\tau}$, …, $\hat{X}_{1,t+L-1+\tau}$—data in the output sequence $\hat{X}_{1,t+\tau}$;

$\hat{X}_{2,t+\tau}$, $\hat{X}_{2,t+1+\tau}$, …, $\hat{X}_{2,t+L-1+\tau}$—data in the output sequence $\hat{X}_{2,t+\tau}$;

$\hat{X}_{N,t+\tau}$, $\hat{X}_{N,t+1+\tau}$, …, $\hat{X}_{N,t+L-1+\tau}$—data in the output sequence $\hat{X}_{N,t+\tau}$;

$X_{1,t}$, $X_{2,t}$, $X_{N,t}$—windows created from the input sequences of Feature 1, Feature 2 and Feature N;

$\hat{X}_{1,t+\tau}$, $\hat{X}_{2,t+\tau}$, $\hat{X}_{N,t+\tau}$—output sequences of Feature 1, Feature 2 and Feature N;

μ, σ—mean and standard deviation of the reconstruction errors;

τ—predicted window length.

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